Financial Data Analysis

Problem Description

The financial analytics platform is studying users' continuous trading performance over a period of time.

You are given an integer array nums, where nums[i] represents the net profit on day i (it may be positive for profit or negative for loss).

The analyst wants to count:

How many continuous time intervals have cumulative profit within the target range [minimum,maximum].

In other words, for any continuous date interval [i, j] (where i <= j), the cumulative profit is defined as:

S(i,j)=nums[i]+nums[i+1]+...+nums[j]

Please compute the number of continuous intervals that satisfy:

minimum<=S(i,j)<=maximum

Input and Output

Input Format

The first line contains two integers: the first is minimum, and the second is maximum.

The second line contains an integer n, representing the length of the array.

The third line contains n integers, representing daily profits.

Output Format

Output one integer, representing the number of valid continuous intervals.

Constraints

- −105≤minimum≤maximum≤105
- 1≤n≤105
- −231≤nums[i]≤231−1

Sample Data

Sample Input 1:

```
-1 3
4
3 -2 4 -5
```

Sample Output 1:

```
5
```

Explanation 1:

[3], [3, -2], [-2, 4], [4, -5], [3, -2, 4, -5]

Sample Input 2:

```
1 3
3
0 1 2
```

Sample Output 2:

```
5
```

Explanation 2:

[1], [2], [0, 1], [1, 2], [0, 1, 2]